

SPACE WEATHER (EP-512)

1. GENERAL

- 1.1 COURSE TITLE: Space Weather
- 1.2 COURSE NUMBER: DE.EP 511.15
- 1.3 CONTACT HRS: 3-0-0, Credit 9
- 1.4 SEMESTER-OFFERED: Odd
- 1.5 PRE-REQUISITE: NIL
- 1.6 SYLLABUS

1. COMMITTEE MEMBER: Dr. A.K. Tripathi (C), BB Karak

2. OBJECTIVE This course is aimed to make students, and engineers in particular, aware of the solar or spatial origin of the disturbances and disruptions that are witnessed in many ground based and spatial technological systems, and to explain the physical mechanisms behind them, to indicate where to find the relevant predictive data and to train how to interpret them in order to lower the risk of damage to technical equipment and economic losses.

3. COURSE CONTENT • The whole set of complex effects of the radiation and the plasma stream from the Sun on the Earth and its magnetosphere; technological systems; climate and the people. (6 Lectures)

- The explosions that occur frequently on the Sun and especially the magnetic plasma clouds (coronal mass ejections); solar drivers of the space weather. (6 Lectures)
- The detectable effects on Earth and its spatial environment in a broad spectrum of time and length scales; various harmful effects for human health and for our technologies. (6 Lectures)
- Polar light (aurora) -- space weather effects; bad weather conditions in space; hindrance or damage satellite operations; disruptions of telecommunication and navigation systems and the related geo-magnetic storms inducing peak currents in pipelines; cause of power grid outages -- a variety of tremendous socio-economic losses. (6 Lectures)
- Solar flares and their effects (4 lectures)
- Radiation risks for the crew and passengers on air planes and astronauts in space; An overview of space weather effects; the physical mechanisms; modeling and prediction. (6 Lectures)

4. READINGS

4.1 TEXTBOOK: 1. R.P. Singhal-2008

Elements of Space Physics, PHI Learning Ltd, N-Delhi

2. Introduction to Space Weather

Mark Moldwin, Cambridge Univ. Press-2021..

4.2 REFERENCE BOOKS: Space Weather Fundamentals, Edited by George V. Khazanov, Taylor & Francis

5. OUTCOME OF THE COURSE: This course gives an overview of space weather effects and the physical mechanisms behind them and explains how these can be modeled and predicted.